

TRUNG V. HA, PhD
Research Assistant, University of Massachusetts Amherst

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Curriculum Vitae

(Last updated: September 11, 2026)

TECHNICAL SKILLS

Programming	Python, C++, MATLAB, MySQL
Packages	NumPy, Pandas, matplotlib, scikit-learn, SciPy, Astropy, JAX
Testing & Deployment	pytest, Sphinx, ReadtheDocs, GitHub Actions
Tools	Linux, Git, Jupyter, VSCode, Slurm/PBS, MPI
ML & Statistics	self-organizing maps, clustering, ensemble methods, hypothesis testing, cross-validation, Monte Carlo resampling, Fourier analysis

RESEARCH PROJECTS

Numerical Simulations of Matter around Supermassive Black Holes

Lead Researcher & Developer

2023 – Present

- Developed numerical solutions in C++ to simulate plasma around supermassive black holes, achieving 100× accuracy improvement against observational data by leveraging novel computational methodologies.
- Ran simulations on distributed supercomputing clusters, leveraging 1000+ CPU cores across MPI-connected networks.
- Secured NSF supercomputing allocation as PI (400,000 CPU hours); managed project timelines and computational resources.
- Analyzed high-dimensional, large-scale ($> 10^7$ data points) simulation outputs (5000+ snapshots) and reported scientific findings to diverse audiences at 3 international conferences, including an invited seminar at the Vietnam National Space Center.
- Publication:
 - “Galaxy- and Meso-scale Multiphase Accretion Flows onto a Supermassive Black Hole I: Gas Morphology and Accretion Dynamics”, **Trung Ha** & Yuan Li, 2026 (manuscript in preparation)

Improving Black Hole – Galaxy Correlation with Machine-learning Regressors

ML Advisor & Manuscript Editor

2024 – Present

- Advised a junior researcher on the implementation of ML algorithms (decision tree, random forest, XG Boost, MLP) to improve prediction of black holes mass from their host galaxies properties by over 2× compared to traditional linear methods.
- Supervised the junior researcher on rigorous ML methodologies, including reproducibility pipeline and convergence tests.
- Played a key role in the manuscript writing process, provided feedback and initiated one-on-one editing sessions.
- Publication:

- “*Black Hole-Host Galaxy Correlations with Machine Learning: A Comparative Study of Illustris, TNG, and EAGLE*”, Jacob Reinheimer, Yuan Li, **Trung Ha**, 2026, The Open Journal of Astrophysics (in review, <https://arxiv.org/abs/2510.20751>)

Anomaly Detection in Plasma Turbulence with Machine Learning

Lead Developer & Analyst

2023 – 2025

- Developed and shipped an open-source Python library, *aweSOM*, that performs unsupervised clustering using self-organizing maps and statistical ensembles, trained on dense, multidimensional datasets ($10^7 - 10^8$ rows).
- Engineered convolutional features enabling novel characterization of two previously unresolved structure classes.
- Refactored and optimized legacy codebases, improving memory utilization by 1000×; implemented JAX for GPU acceleration, achieving up to 100× speedup over CPU baseline.
- Designed scaling/unit tests, documented and deployed on ReadtheDocs; adopted by external groups as a production tool.
- Delivered 4 presentations at international conferences in the US and Europe and an invited departmental seminar.
- Publications:
 - “*Machine Learning Characterization of Intermittency in Relativistic Pair Plasma Turbulence: Single and Double Sheet Structures*”, **Trung Ha** et al. 2025, Astrophysical Journal Letters (impact factor: 11.7), 985, 31 (<https://iopscience.iop.org/article/10.3847/2041-8213/add47b>), 5 citations
 - “*aweSOM: a CPU/GPU-accelerated Self-organizing Map and Statistically Combined Ensemble Framework for Machine-learning Clustering Analysis*”, **Trung Ha** et al. 2025, Journal of Open Source Software, 10(108), 7613 (<https://joss.theoj.org/papers/10.21105/joss.07613>)

Novel Characterization of Weak Line Quasars’ Properties

Lead Analyst & Scientific Coordinator

2020 – 2023

- Led an international team of 22 scientists from five countries (USA, Canada, China, Chile, Israel), synthesized ideas among experts into a unified analysis framework.
- Developed Python analysis pipelines to extract 20+ properties from spectral observations of hundreds of quasars.
- Migrated legacy MATLAB codebase to Python; reproducible workflow subsequently adopted as group standard.
- Reported novel findings of a non-linear correlation between quasars’ accretion rate and their UV spectral signal, disproving a widely assumed break in quasars classification.
- Publications:
 - “*Shedding New Light on Weak Emission-line Quasars in the C IV–H β Parameter Space*”, **Trung Ha** et al. 2023, Astrophysical Journal (IF: 5.4), 950, 97 (<https://iopscience.iop.org/article/10.3847/1538-4357/acd04d>), 10 citations
 - “*Gemini Near Infrared Spectrograph–Distant Quasar Survey: Rest-frame Ultraviolet–Optical Spectral Properties of Broad Absorption Line Quasars*”, Harum Ahmed, ..., **Trung Ha** et al.

2024, Astrophysical Journal (IF: 5.4), 968, 77 (<https://iopscience.iop.org/article/10.3847/1538-4357/ad3e69>), 7 citations

Large-scale Statistical Inference on Multi-dimensional Observational Data

Lead Researcher

2020 – 2022

- Built Pandas ETL pipelines to clean, transform, and validate data integrity across millions of observations.
- Implemented resampling methods (Monte Carlo, jackknife, bootstrap) to estimate systematic uncertainties in noisy data.
- Developed a parallel two-point correlation estimator achieving 20× higher computational throughput than existing codebases.
- Delivered 3 presentations in the US and Asia and was interviewed by the American Astronomical Society on publication.
- Publications:
 - “*Turbulence in Milky Way Star-forming Regions Traced by Young Stars and Gas*”, **Trung Ha** et al. 2022, Astrophysical Journal (IF: 5.4), 934, 7 (<https://iopscience.iop.org/article/10.3847/1538-4357/ac76bf>), 41 citations
 - “*Measuring Turbulence with Young Stars in the Orion Complex*”, **Trung Ha** et al. 2021, Astrophysical Journal Letters (IF: 11.7), 907, 40 (<https://iopscience.iop.org/article/10.3847/2041-8213/abd8c9>), 23 citations

Scientific Mentor

2024 – 2025

- Mentored a post-baccalaureate student from zero domain knowledge to first-author publication on multi-scale correlation functions and MCMC simulations of stellar data, expanding coverage of proposed methodology from early works.
- Publication:
 - “*Probing Turbulence, Gravity, Supernovae, and Magnetic Field Effects with the 6D Kinematics of Young Stars in Milky Way Star-forming Regions*”, Benjamin Velguth, Yuan Li, **Trung Ha** et al. 2025, Astrophysical Journal (IF: 5.4), 990, 165 (<https://iopscience.iop.org/article/10.3847/1538-4357/adf5c0>)

WORK & RESEARCH EXPERIENCE

Sep 2024 – Present	Graduate Research Assistant , University of Massachusetts Amherst
Jun 2021 – Aug 2024	Graduate Research Assistant , University of North Texas
Sep 2023 – May 2024	Research Analyst (Pre-doctoral Fellow) and Guest Researcher , Center for Computational Astrophysics, Simons Foundation
Aug 2020 – May 2021	Graduate Teaching Assistant , University of North Texas
Sep 2018 – May 2020	Undergraduate Research Student , Center for Computational Relativity and Gravitation, Rochester Institute of Technology
Sep 2018 – Dec 2019	Undergraduate Teaching Assistant , University of Rochester
Jun 2018 – Aug 2018	Summer Research Intern , Laboratory for Laser Energetics, University of Rochester

EDUCATION

2024 – 2026	University of Massachusetts Amherst, Amherst, Massachusetts PhD in Astronomy – conferred September 2026 Thesis title: “ <i>The Next Generation of Diagnostic Tools for Astrophysical Turbulence</i> ”, advisor: Yuan Li GPA: 4.00 / 4.00
2020 – 2024	University of North Texas, Denton, Texas PhD candidate – transferred to UMass before receiving degree Master of Science in Physics – conferred May 2022 GPA: 4.00 / 4.00
2017 – 2020	University of Rochester, Rochester, New York Bachelor of Science in Physics

SELECTED TALKS

Dec 2025	Stars & Plasma Astrophysics Group Meeting , Flatiron Institute, New York, NY, USA Title: “Modeling Multiphase Accretion Flows onto M87* with Galaxy- and Meso-scale Simulations”
Jun 2025	INVITED - Vietnam National Space Center , Astrophysics Department Seminar, Hanoi, Vietnam Title: “Feedback-driven Multiphase Accretion in M87: Results from Meso-scale Simulations”
Jun 2025	IAUS 397 – UniversAI: Exploring the Universe with Artificial Intelligence , Athens, Greece Title: “aweSOM: an Open-source Python Package for Efficient Clustering of Intermittency in Magnetized Plasma Turbulence”
Mar 2025	Stars & Plasma Astrophysics Group Meeting , Flatiron Institute, New York, NY, USA Title: “Segmentation of Single and Double Current Sheets in Magnetized Plasma Turbulence with Machine Learning”
Jun 2024	AstroAI Workshop , Harvard-Smithsonian Center for Astrophysics, Cambridge, MA, USA, and Midwest Magnetic Fields Workshop , Madison, WI, USA Title: “Segmentation of Current Sheets in Magnetized Plasma Turbulence with Computer Vision”
Apr 2024	INVITED - Center for Computational Relativity and Gravitation Lunch Talk , Rochester, NY, USA Title: “Can Neural Networks Recognize Current Sheets? Using Computer Vision to Analyze Magnetized Plasma Turbulence”
Feb 2024	Kavli Institute for Theoretical Physics (KITP) – Turbulence in the Universe Workshop , Santa Barbara, CA, USA Title: “Segmentation of Current Sheets in Magnetized Plasma Turbulence with Computer Vision”
Dec 2023	Black Holes on Broadway: The Next Generation of AGN Models in Galaxy Formation , New York, NY, USA Title: “Bridging the Gap: Modeling Supermassive Black Holes Feeding and Feedback at the Meso-scale”
Aug 2022	Star Formation in Different Environments 2022 , Rencontres du Vietnam, Quy Nhon, Vietnam

Title: “Turbulence in Milky Way Star-forming Regions Traced by Young Stars and Gas”

PROFESSIONAL SERVICE

Served as referee for the Astrophysical Journal Letters.

Mentored two undergraduate students (Alexis NieMeira, Jonathan Red), one post-baccalaureate fellow (Benjamin Velguth), and two junior graduate students (Jacob Reinheimer, Roy Hsu), resulting in two publications (Velguth et al. 2025 and Reinheimer et al. 2026, as third author).

SUPERCOMPUTING & OBSERVING AWARDS

Dec 2024	Co-I., XMM-Newton proposal Title: “Identifying a Robust and Practical Accretion-Rate Indicator for Distant Quasars”; observing time: 110 kiloseconds.
Mar 2024	P.I., NSF ACCESS Explore allocation Title: “Turbulent multiphase accretion flows from supermassive black hole feedback”; amount: 400,000 CPU hours

AWARDS AND HONORS

Spring 2025	Mary Dailey Irvine Graduate Travel Grant, UMass Amherst, \$1000
2023 – 2024	The Zhibing Hu Scholarship, University of North Texas, \$1000.
May 2023	Featured on University of North Texas College of Science news: “UNT Physics Graduate Student Selected for Prestigious Pre-Doctoral Program” (link to article)
Spring 2023	College of Science Travel Award, University of North Texas. \$500.
Feb 2022	Featured on University of North Texas’s Graduate Research Spotlight .
2021 – 2025	R. B. Toulouse Scholarship, University of North Texas. \$1000/year.
Apr 2021	Featured on the <i>North Texan</i> : “Turbulent Motion Moves Research Forward” (link to article).
2019 – 2020	Take Five Scholar, University of Rochester. Thesis: “Exploring the Advantages and Shortcomings of French Literature in Translation”.

OTHER ACTIVITIES

Participated in the Flatiron Institute’s Center for Computational Astrophysics Pre-doctoral program in New York City in fall 2023.

Organizer for the weekly joint-UNT/UTD astronomy journal club, 2023.

Interviewed on AAS Journal Author Series about publication titled: “Measuring Turbulence with Young Stars in the Orion Complex” on [YouTube](#), 2021.

Participated in student exchange programs: “Cultural Exchange Program” in Arizona, USA during the 2014-15 school year and “French in France” in Rennes, France in summer 2019.

Citizenship: Vietnam

U.S. Permanent Resident

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Publication List

In reverse chronological order, refereed publications are listed first

FIRST AUTHOR PUBLICATIONS

1. ***“Machine-Learning Characterization of Intermittency in Relativistic Pair Plasma Turbulence: Single and Double Sheet Structures”***
Ha, Trung; Nättälä, J.; Davelaar, J.; Sironi, L. ([2025ApJ...985L..31H](#))
2. ***“aweSOM: a CPU/GPU-accelerated Self-organizing Map and Statistically Combined Ensemble Framework for Machine-learning Clustering Analysis”***
Ha, Trung; Nättälä, J.; Davelaar, J. ([2025JOSS.07613](#), [GitHub](#))
3. ***“Shedding New Light on Weak Emission-Line Quasars in the CIV–H β Parameter Space”***
Ha, Trung; Dix, C.; Matthews, B. M.; Shemmer, O.; et al., ([2023ApJ...950...97H](#))
4. ***“Turbulence in Milky Way Star-forming Regions Traced by Young Stars and Gas”***
Ha, Trung; Li, Y.; Kounkel, M.; Xu, S.; Li, H.; Zheng, Y., ([2022ApJ...934....7H](#))
5. ***“Measuring Turbulence with Young Stars in the Orion Complex”***
Ha, Trung; Li, Y.; Xu, S.; Kounkel, M.; Li, H., ([2021ApJ...907L..40H](#))
6. ***“Galaxy- and Meso-scale Multiphase Accretion Flows onto a Supermassive Black Hole I: Gas Morphology and Accretion Dynamics”***
Ha, Trung; Li, Y., (in prep.)

OTHER PUBLICATIONS

7. ***“Probing Turbulence, Gravity, Supernovae, and Magnetic Field Effects with the 6D Kinematics of Young Stars in Milky Way Star-forming Regions”***
Velguth, B.; Li, Y.; Ha, Trung, et al. ([2025ApJ...990..165V](#))
8. ***“Rest-Frame Optical Spectroscopy of Ten $z \sim 2$ Weak Emission-Line Quasars”***
Chen, Y.; ...; Ha, Trung, et al. ([2024ApJ...972..191C](#))
9. ***“Gemini Near Infrared Spectrograph - Distant Quasar Survey: Rest-Frame Ultraviolet-Optical Spectral Properties of Broad Absorption Line Quasars”***
Ahmed, H.; ...; Ha, Trung, et al., ([2024ApJ...968...77A](#))
10. ***“The Nature of the Motions of Multiphase Filaments in the Centers of Galaxy Clusters”***
Ganguly, S.; ...; Ha, Trung, ([2023FrASS..1038613](#))
11. ***“Handing-Off the Outcome of Binary Neutron Star Mergers for Accurate and Long-Term Post-Merger Simulations”***
Lopez Armengol, F. G.; ...; Ha, Trung; et al., ([2022PhRvD.106h3015L](#))
12. ***“HARM3D+NUC: A new method for simulating the post-merger phase of binary neutron star mergers with GRMHD, tabulated EOS and neutrino leakage”***
Murguia-Berthier, A.; ...; Ha, Trung, et al., ([2021ApJ...919...95M](#))
13. ***“Black Hole-Host Galaxy Correlations with Machine Learning: A Comparative Study of Illustris, TNG, and EAGLE”***
Reinheimer, J.; Li, Y.; Ha, Trung, et al. (submitted to OJA, [arXiv:2510.20751](#))